

# ANDREAS BOSCARIOL

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## EDUCATION

### The University of British Columbia

*Bachelor of Applied Science, Major in Chemical Engineering*

**Vancouver, Canada**

Class of May 2027

## EXTRACURRICULARS

### UBC Trading Group

*Factor Research Lead*

**Vancouver, Canada**

Oct 2023 - Present

- Leading a team of 5 to develop a GAN model using CNN and LSTM architectures in PyTorch to model stock returns, overseeing model design and training phases
- Applying reinforcement learning techniques to optimize hyperparameters, aiming to enhance prediction accuracy and generalize effectively to unseen data
- Performing feature extraction with various models (RFE, XGBoost, PCA, etc.) to reduce dimensionality and mitigate overfitting, refining the model's ability to capture complex patterns in stock data

*Software Developer*

Oct 2022 - Oct 2023

- Managed the UBC Trading Group's financial database with Python and SQL, increasing data coverage by integrating external datasets on individual stocks and macroeconomic indicators.
- Automated data acquisition using WRDS and BEA APIs, reducing manual processing time and ensuring up-to-date market insights.

### UBC Thunder Bikes

*Electrical Team*

**Vancouver, Canada**

Sept 2024 - Present

- Modeled battery components and assemblies in SolidWorks, accelerating prototyping of a high-performance electric racing motorcycle.
- Conducted BMS testing and manufactured bus bars via water jetting.
- Developed an interactive [website](#) with a responsive design, boosting the PageSpeed Insights performance score from 28 to 100
- Created sponsorship materials and performed outreach to secure partnerships

### UBC Uncrewed Aircraft Systems

*Software Developer*

**Vancouver, Canada**

Sept 2023 - May 2024

- Programmed collision detection algorithms in C++ using LiDAR data for object detection and classification for drones
- Applied noise reduction and downsampling to point clouds, enhancing data quality for obstacle identification
- Implemented RANSAC and clustering algorithms to segment and group data, increasing the accuracy of collision avoidance systems
- Developed algorithms for bounding box generation around obstacles, aiding in spatial awareness and navigation safety

## PROJECTS

### Website Portfolio ([andreasboscariol.com](https://andreasboscariol.com))

May 2024

- Designed and developed a personal website using ReactJS and CSS, incorporating dynamic, draggable, and resizable components for an interactive user experience

### Cinema Critic ([cinemacritic.xyz](https://cinemacritic.xyz))

June 2024

- Developed a full-stack web application for critiquing user movie reviews using HTML, CSS, and JavaScript
- Leveraged AWS Amplify for deployment and used AWS Lambda for backend tasks, including fetching and parsing RSS feeds from Letterboxd and integrating ChatGPT API for personalized feedback

## ADDITIONAL

Skills: Solidworks, C++, Java, Python, 3D Printing, DBeaver  
Interests: Hackathons, CTFs, Finance, Reading, Movies